Two-Phase Positioning System Based on the Fusion of Wi-Fi Signal Strength and Pose Estimation

Han-Hsuan Cheng, Jenq-Shiou Leu, *Senior Member, IEEE*, and Jin-Xian Liu

***Abstract*—Due to the growing importance of positioning in recent years, global positioning system has become widely used. However, indoor positioning presents significant challenges due to multipath interference, which can cause large signal fluctuations and high positioning errors for signal-based approaches. Recent advances in deep learning have enabled researchers to estimate people's positions and locate them indoors using image recognition technology, but identification of individuals remains challenging. Therefore, we propose a highly accurate indoor positioning approach using Wi-Fi signals and images. This research employs smartphones to measure the signal strength of three Wi-Fi base stations' 2.4 and 5 GHz frequency bands. We then use machine learning techniques to estimate individuals' positions with coarse accuracy. Subsequently, we analyze the positions of individuals in surveillance camera images. We use OpenPose, an open-source multiperson pose estimation system, and propose a foot position extraction algorithm to extract the foot positions of individuals in the image. We obtain the coordinates of the people in the image through direct linear transformation and machine learning. Finally, we match the coordinates obtained from the two techniques to identify the user's device. Experimental results show that the proposed method achieves higher accuracy, with an average error of 0.43 meters.**

***Index Terms*—Artificial intelligence, global positioning system (GPS), image recognition, machine learning, wireless communication.**

I. Introduction

PEOPLE have been relying on global positioning system (GPS) [1] technology for positioning since the 1970s. New services leveraging GPS technology, such as Uber Eats, dating software, and other high-value apps that played a significant role during the coronavirus outbreak, have emerged alongside the widespread use of smartphone applications.

Although GPS can achieve high accuracy outdoors, its signal strength is often attenuated due to the sheltering effect of buildings, making indoor positioning a challenge. As a result, overcoming the indoor positioning issue has become an important problem in the relevant research area. Indoor positioning can be incorporated into a wide variety of applications. For instance, in shopping malls, merchants can offer personalized product recommendations and discounts to customers based on their location; hospitals can track the flow of patients using positioning; and factories can manage the movement of people and oversee hazardous areas. Positioning presents numerous business applications and possibilities.

Regarding positioning, traditional methods often use fingerprinting-based algorithms to locate users by collecting Wi-Fi or Bluetooth signal strength in buildings. However, accuracy can be greatly affected by signal fluctuations and interference from obstacles. In contrast to conventional positioning methods, a deep learning-based ultrawideband (UWB) localization approach [2] achieves higher accuracy by using a deep long short-term memory network to predict the user's position based on anchor-tag distance information. Nevertheless, deploying UWB requires dedicated infrastructure, which can be challenging in complex building structures, and external hardware (i.e., tags), which limits scalability.

Instead of relying on the infrastructure of buildings, Cosma et al. [3] proposed an image-based positioning solution using deep neural networks on fixed cameras with limited computing resources. They claim that high-precision real-time positioning can be achieved. Another promising approach is pedestrian dead reckoning (PDR). Zhao et al. [4] implemented a 2-D PDR system using smartphones and enhanced a particle filter to prevent the wall-crossing problem. However, the disadvantage of PDR is the accumulation of location errors over time, particularly during turns and heading drifts. To achieve better stability and higher accuracy, Hao et al. [5] developed a cross-layer positioning method based on multi-sensor and Wi-Fi fingerprint fusion. They utilized the unscented Kalman filter to combine Wi-Fi and PDR positioning results, outputting the best two-dimensional estimation.

IHS Markit [6] predicts that the global number of surveillance cameras will increase by over 30% in the next two years, from its current level of 770 million. In the United States, the average number of surveillance cameras per person is 4.1. Consequently, this article proposes a two-stage positioning system that utilizes existing Wi-Fi and image data to achieve real-time, high-accuracy positioning without the need for additional equipment. This system can be customized to suit the environment and enhance the accuracy of identifying individuals.

To design an indoor positioning system, it is important to consider diverse environments and specific needs, as technology choices may vary. There is no universal solution that can address

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The authors are with the Department of Electronic and Computer Engineering, National Taiwan University of Science and Technology, Taipei 106216, Taiwan (e-mail: m10802156@mail.ntust.edu.tw; jsleu@mail.ntust.edu.tw; m11002135@mail.ntust.edu.tw).

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all situations. Unique positioning methods have emerged, with wireless signal and visual image positioning being dominant. Wireless-based systems use sensors, such as Bluetooth, Wi-Fi, and UWB to calculate distances through time difference, signal strength, and angle, using a signal attenuation model for absolute positioning. On the other hand, vision-based systems use pre-collected image data and deep learning models to recognize objects or people, determining positions based on surrounding images and pedestrian location estimation.

The main contributions of this article are as follows.

1) Developing a two-phase positioning system based on Wi-Fi signal strength and pose estimation contributes to the accurate identification of people.
2) Using machine learning models trained on combined 2.4 GHz and 5 GHz received signal strength indicator (RSSI) as features contributes to reducing Wi-Fi positioning error in complex environments.
3) Enhancing the human foot position extraction algorithm using the pose estimation model and 2-D direct linear transformation (2-D-DLT) contributes to real-time, high-accuracy positioning with an average error of within 1 meter.

The rest of this article is organized as follows. The next section provides a brief background and a summary of recent studies on indoor positioning. Section III gives a brief overview of the system architecture, and proposed two-phase positioning methods. Section IV evaluates the performance of different indoor positioning methods and the experiment results. Finally, Section V concludes this article.

II. Related Work

This article proposes a system that combines image-based and Wi-Fi-based indoor positioning methods, including Wi-Fi signal strength positioning and pedestrian location estimation technology. In this section, we discuss relevant methodologies and current research on indoor positioning.

According to [7], the challenges of indoor positioning include the multipath effect, blocking obstacles, and device heterogeneity, which can reduce the accuracy of Wi-Fi-based indoor positioning systems. Muthukrishnan et al. [8] shows that the error rate of image-based indoor positioning technology is lower than that of other technologies. However, image-based indoor positioning has difficulty identifying users' devices. Consequently, fusion indoor positioning technology has been proposed in many studies.

Liu et al. [9] proposed the VMag system, a hybrid fusion method that combines the received signal strength of geomagnetic fields and visual images due to their complementary nature [10]. The system does not require any infrastructure; it simply needs users holding smartphones. The system performs well in four typical indoor settings: a research laboratory, a garage, a cafeteria, and an office, with 91% of localization errors falling within 0.85, 1.34, 1.34, and 0.85 m, respectively.

Poulose et al. [11] proposed a sensor fusion framework that uses PDR and Wi-Fi localization systems, which shows better indoor positioning accuracy than traditional Wi-Fi positioning systems. The proposed framework achieves a maximum localization error of 1.17 m for the rectangular motion of a pedestrian and 0.44 m for linear motion.

Pedestrian location estimation [12] is a vision-based indoor positioning system that relies on pose estimation or object detection. This is usually achieved by placing a camera in a fixed position to capture a bird's-eye view image. These images are then analyzed using a well-developed object detection model, such as RCNN or Yolo [13], [14], and a pose estimation model, such as OpenPose [15], to detect the joints of the human body in the image. The system extracts the pixel coordinates near the foot position and converts them to real-world coordinates using direct linear transformation. The distance error of pedestrian location estimation is typically within 1 m. However, this error may increase due to lens distortion or occlusion of human joints, which can result in incorrect foot position identification.

In the past, multiple-person real-time 2-D pose estimation was challenging because each image contained an unknown number of individuals in various positions, and limb contact or overlap created complicated spatial interference. These issues made joint point connections more difficult and increased computational complexity as the number of people grew. Poulose et al. [16] proposed OpenPose, a multiperson real-time pose estimation system that can differentiate each human body and its joint positions amidst the complex background of multiple individuals and identify the streaming image.

Based on the above overview of various current indoor localization techniques and pose estimation, we propose an indoor positioning system that combines pose estimation and Wi-Fi signal strength. We use OpenPose for pose estimation, and we propose a foot position extraction algorithm based on joints to complete a high-accuracy positioning system using direct linear transformation and machine learning techniques. In addition, the system can determine the user's identity by matching two coordinates obtained from the two techniques.

III. Methodology

In this section, we provide an overview of the proposed indoor positioning system by describing its architecture, as well as the image-based and Wi-Fi-based techniques employed. Additionally, we discuss the fusion method used to combine these approaches, which enables user device identification and high positioning accuracy.

A. Description of the Indoor Positioning System

Leveraging the ubiquity of Wi-Fi base stations and surveillance cameras in daily life, we propose an indoor positioning system. The smartphone receives Wi-Fi signal strength and, in conjunction with the fixed-point field monitoring from the surveillance camera, estimates the human pose within the image. By employing machine learning techniques, we can estimate people's positions more accurately, deducing the individual's location based on Wi-Fi and image data, and ultimately matching them to identify the user.

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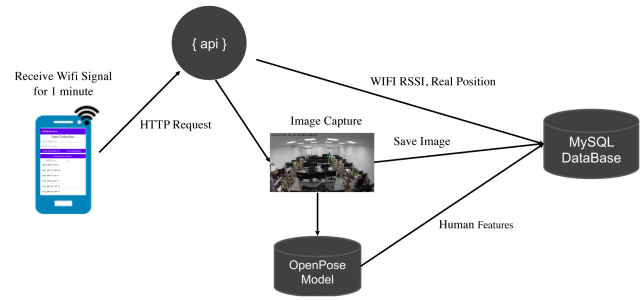


Fig. 1. Architecture of the offline phase.

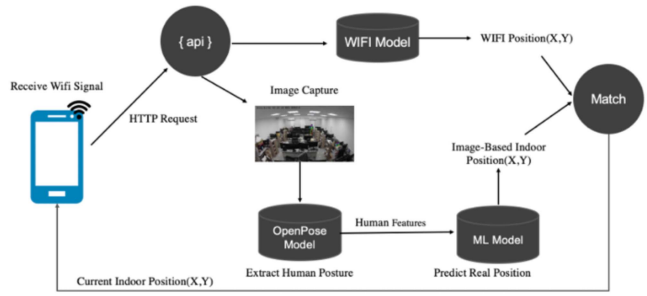


Fig. 2. Architecture of the online phase.

B. System Architecture

The operation of the system is divided into offline and online phases. The offline phase involves gathering Wi-Fi signals and performing pose estimation of human characteristics, whereas the online phase utilizes a constructed machine learning model to predict human coordinates.

1) *Offline Phase:* The architecture of this phase is shown in Fig. 1. In advance, we placed 3 Wi-Fi access points in the field and established 75 reference positions. We collect data 12 times for every position (every 5 s for a total of 1 min) in four directions and send a request containing the Wi-Fi strength and the current coordinates to the offline system. The system connects the camera to capture an image at the same time and extracts the position of human joints in the image using the OpenPose model. Finally, save the results to the database.
2) *Online Phase:* The architecture of this phase is shown in Fig. 2. The system uses the Wi-Fi-based positioning machine learning model to predict the coordinates based on the signal strength. Simultaneously, the camera captures an image, and the coordinates of all people are obtained by using the proposed foot position extraction algorithm and the OpenPose model. Finally, we use the matching method to match the coordinates generated by both ways to identify the user, which is described next.

C. Matching Method

Using the image and Wi-Fi signal strength, we derive the coordinates using the methods proposed in this article.

The matching method identifies the user by considering the coordinates of the person closest to the coordinates inferred from the Wi-Fi RSSI. The distance between two points is calculated using the Euclidean distance. An example of the concept is shown in Fig. 3.

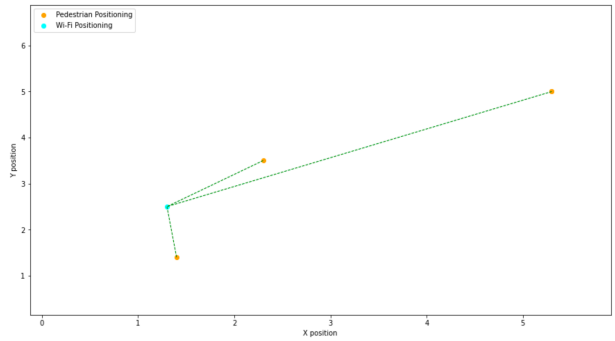


Fig. 3. Example of matching coordinates from Wi-Fi-based and image-based positioning methods.

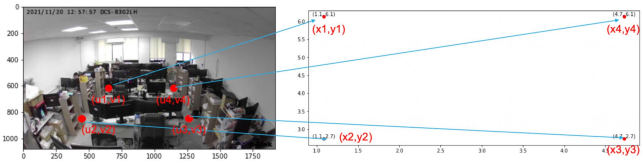


Fig. 4. Coordinates after performing direct linear transformation.

D. Coordinates Transformation

Direct linear transformation [17], also known as DLT, is a concept based on the idea that the coordinates of a person's foot position in an image can be determined after obtaining the foot position of the person in the image. To convert the coordinates in the image to the coordinates in the real field, the coordinates from the camera image need to be transformed into coordinates from a bird's-eye view.

Direct linear transformation [17] can transform object-space coordinate$(X_i,\ Y_i,\ Z_i)$ into image-plane coordinates $(u_i,\ v_i)$. In this system, both camera image coordinates and aerial view coordinates must be considered. After obtaining four pairs of coordinates between the camera image $(u_i,\ v_i)$ and the bird's-eye view $(X_i,\ Y_i,\ Z_i)$, the planar homography matrix H may be obtained by solving the following equation:

$$\begin{bmatrix} X_i \\ Y_i \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} \tag{1}$$

After obtaining the matrix H, the following equations are used to find the relationship between the camera image and the bird's-eye view. Fig. 4 shows the result of the 2-D-DLT

$$X = \frac{h_{11}u + h_{12}v + h_{13}}{h_{31}u + h_{32}v + 1} \tag{2}$$

$$Y = \frac{h_{21}u + h_{22}v + h_{23}}{h_{31}u + h_{32}v + 1}. \tag{3}$$

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Fig. 5. Image of a person with obscured joints.

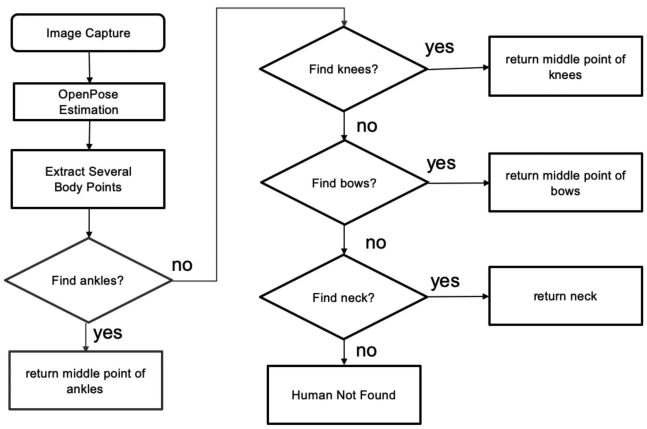


Fig. 6. Process flowchart of the foot position extraction algorithm.

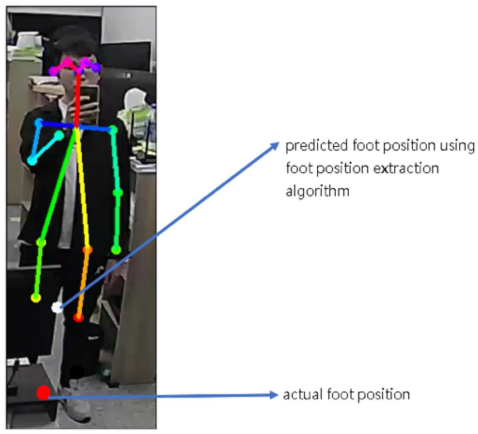


Fig. 7. Position after using the foot position algorithm.

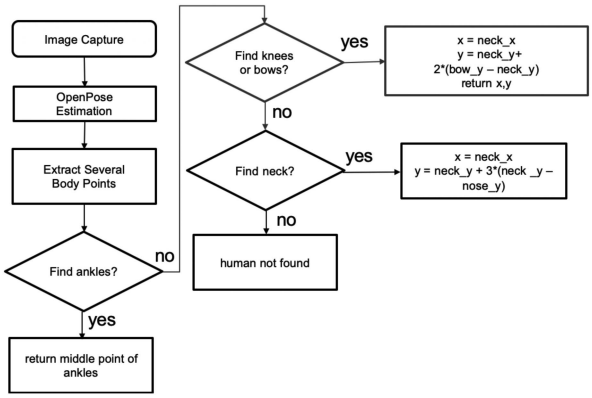


Fig. 8. Process flowchart of the improved foot position extraction algorithm.

E. Foot Position Extraction Algorithm

Before describing the foot position extraction algorithm, we need to understand the description of the COCO [18] dataset. COCO [18] is a large open-source dataset developed by Microsoft, Facebook, CVDF, and other organizations. It contains 330 000 images and 1.5 million objects organized into 80 categories, and is commonly used in computer vision. OpenPose processes the COCO data and generates 18 joints.

As shown in Fig. 5, the body is obscured, resulting in an inadequate number of detected joints when using OpenPose to identify the human body. In some cases, the ankle may not even be detected. To address this issue, a foot position extraction algorithm was developed. The processing flow for this algorithm can be seen in Fig. 6.

First, we employ the OpenPose model for pose estimation to obtain the body joints from the image. At this stage, it is possible to determine whether the ankle joints can be found. If so, the algorithm returns the midpoint of the joints directly. From bottom to top, it checks if the knees, hips, and neck can be detected. If any of these joints are found, the algorithm returns the midpoint of the corresponding joint. If, in the end, the neck cannot be found, the algorithm returns a result indicating insufficient information about the person's body.

By using 2-D-DLT, we can transform the image pixels into actual field coordinates. However, these coordinates may be affected by the deviation of the image pixel points, which can in turn impact the actual position projection. In the foot position extraction algorithm, the calculated foot position and the actual foot position can differ significantly in cases of occlusion.

Fig. 7 illustrates that most of the right foot is not visible. Using the foot position extraction algorithm shown in Fig. 6 would result in the extracted foot position located at the midpoint of the knees. However, this would cause a significant error after performing 2-D-DLT. As a result, this research proposes an improved algorithm for extracting foot positions. The improved algorithm's process flowchart is shown in Fig. 8. Since the distance from the hip to the neck is similar to the distance from the hip to the foot, this algorithm can approximate the foot position. Fig. 9 shows the calculated foot position using both the original and improved algorithms, and the results demonstrate that the improved algorithm provides a closer approximation to the actual foot position.

In cases where only center point of the body is identified and the hip or knee joints remain undetected, an alternative method for estimating foot position must be employed. The X coordinates of the foot position continue to be based on the X coordinates of the neck, while the Y coordinates of the foot position are extrapolated downward from the neck by three times the distance between the nose and the neck. The results, as illustrated in Fig. 10, demonstrate that the foot position predicted

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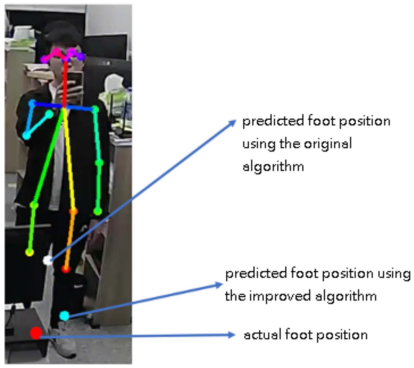


Fig. 9. Positions after using the improved foot position algorithm compared with the unimproved foot position algorithm (1).

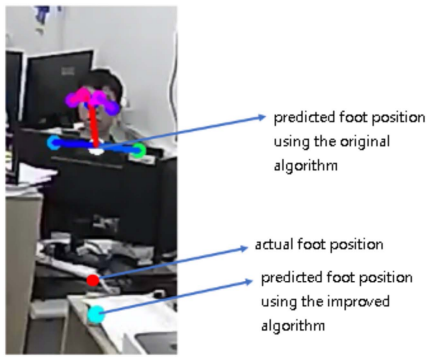


Fig. 10. Positions after using the improved foot position algorithm compared with the unimproved foot position algorithm (2).

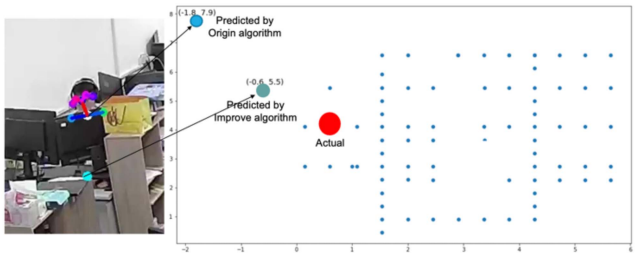


Fig. 11. Coordinates of the improved foot position extraction algorithm and the unimproved algorithm after performing direct linear transformation.

by the improved algorithm is significantly closer to the actual foot position.

Fig. 11 demonstrates that the improved coordinates are evidently closer to the actual coordinates. Nevertheless, lens distortion and coordinate transformation introduce an error of nearly one meter. Thus, we can utilize classical regression machine learning models to establish a relationship between the coordinates derived from the 2-D-DLT and the actual coordinates, aiming to minimize the distance error.

F. Enhancing Image-Based Positioning Using Machine Learning

Due to the lens deviation of the camera and the foot position error produced by the pose estimation model, the transformed coordinates obtained using the 2-D-DLT approach deviate significantly from the real coordinates. Therefore, on the basis of 2-D-DLT, we extract important features to train classic regression models to reduce errors. We employ models, such as linear regression [19], Bayesian ridge [20], and gradient boosting regression [21]. These models incorporate features as given in Table I.

TABLE I
EXAMPLE OF FEATURES FOR IMAGE-BASED POSITIONING MACHINE LEARNING MODEL

Number of Detected joints	Foot position x(pixel)	Foot position y(pixel)	Foot position x(m) after 2-D-DLT.	Foot position y(m) after 2-D-DLT.
Ex: 8	602	713	2.8	3.4

TABLE II
EXAMPLE OF FEATURES FOR WI-FI-BASED POSITIONING MACHINE LEARNING MODEL

Position (x, y)	Average Wi-Fi RSSI 2.4GHz (Ap1~Ap3)	Average Wi-Fi RSSI 5GHz (Ap1~Ap3)
Ex: (0, 0)	-38.58, -39.4, -36.57	-33.58, -32.3, -39.57

G. Proposed Wi-Fi-Based Positioning Method

Most routers operate in the 2.4 and 5 GHz bands. Previous studies have only used 2.4 GHz for Wi-Fi positioning using RSSI. However, 5 GHz provides better positioning results and avoids interference from other devices operating in the 2.4 GHz band [22]. Therefore, we collected signal strengths at both 2.4 GHz and 5 GHz to obtain diverse features.

Due to the fluctuation of signal strength caused by various factors such as physical obstructions, signal sharing, and other electrical devices, we collected 12 RSSI records at every position, totaling 900 records, and processed the data for each position by taking the average of signal strength values. The extracted features are given in Table II. We use classic machine learning models for accurate Wi-Fi-based positioning.

IV. EXPERIMENTS

In the experiment, an area measuring 5.8 m in length and 4.3 m in width was chosen as the experimental field. We will employ the proposed methods from Section III, along with classic machine learning models, to enhance the accuracy of indoor positioning. Each method will be compared using a visualization tool.

A. Data Collection

Data was collected from 75 positions, as illustrated in Fig. 12. Three Wi-Fi base stations were deployed to serve as sources of Wi-Fi signal strength. To gather Wi-Fi RSSI from various angles, measurements were taken from the front, rear, left, and right at each position, as well as from both 2.4 and 5 GHz frequency bands. A total of 12 Wi-Fi signal strength data points were

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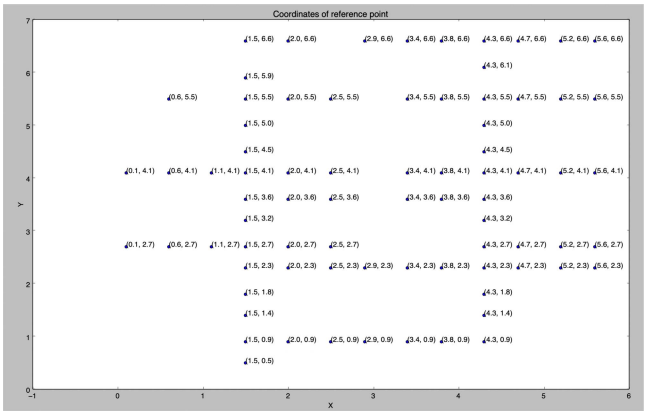


Fig. 12. Coordinates of the reference points.

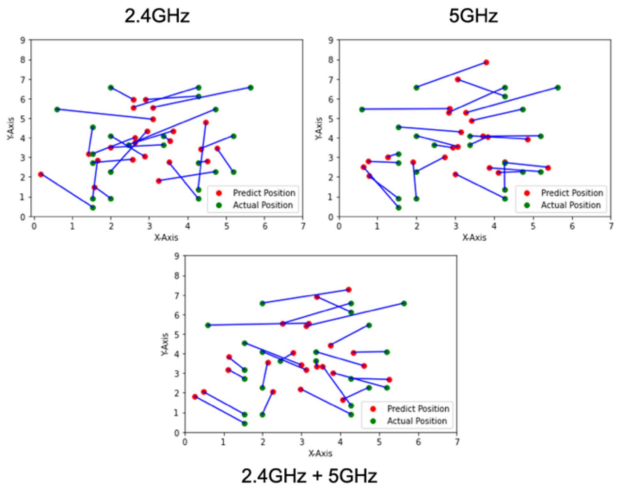


Fig. 13. Visual representation of the predicted and actual coordinates for each Wi-Fi frequency.

TABLE III
DISTANCE ERROR OF AVERAGE OF RSSI

Model	2.4GHz	5GHz	2.4GHz + 5GHz
Linear Regression [19]	1.71m	1.51m	1.47m
Bayesian Ridge [20]	1.76m	1.51m	**1.46m**
Gradient Boosting Regression [21]	2.00m	1.72m	1.68m

collected from each position within a one-minute timeframe. The surveillance camera captured 75 images with person standing at different reference points, including 52 images where the person was partially obscured.

B. Results of Wi-Fi-Based Indoor Positioning

We trained machine learning models using three classic regression models and divided the data into training and test datasets at a ratio of 8:2. The results are presented in Table III. The error in the 2.4 GHz band is greater than that in the 5 GHz band due to numerous interferences. By using both the 2.4 and 5 GHz bands simultaneously, the distance error might be reduced significantly. Among the models, the Bayesian Ridge yielded the most accurate results with an error of 1.46 m. To provide a better understanding of the error statistics for this method, we included the standard deviation and median, which are 0.64 and 1.42 m, respectively.

To showcase the performance of each method, we illustrated the difference between predicted and actual coordinates, displayed the model prediction outcomes in three distinct frequency bands, and applied color-coding to the actual positions. As demonstrated in Fig. 13, the distance error within the 2.4 GHz band is substantial, resulting in a relatively irregular distribution range of predicted coordinates. Conversely, in the 5 GHz band and the combined 2.4 and 5 GHz bands, predicted coordinates frequently align with the actual coordinates, with the distribution closely matching the real positions. This finding also suggests that employing higher-frequency band signals as features can contribute to a reduction in distance error.

TABLE IV
DISTANCE ERROR OF A MATH-ONLY MODEL AND MACHINE LEARNING MODELS

Method	Model Type	Original Foot Position Algorithm Distance Error	Improved Foot Position Algorithm Distance Error
2D-DLT	Math-Only Model (Baseline)	2.25m	0.84m
2D-DLT and Linear Regression [19]	Machine Learning Model (Proposed Method)	0.82m	**0.43m**
2D-DLT and Bayesian Ridge [20]		0.92m	0.46m
2D-DLT and Gradient Boosting Regressor [21]		0.83m	0.51m

C. Results of Image-Based Indoor Positioning

The experimental results can be given in Table IV, with the error value representing the average distance error between the predicted coordinates and the true coordinates. In order to compute this mean distance error, we divided the data into five subsets and applied the k-fold cross-validation method for model evaluation.

The findings from the original foot position extraction algorithm suggest that the implementation of machine learning models has the potential to significantly decrease the average distance errors in comparison to the baseline model. The optimal performance was achieved through linear regression, resulting in an error of 0.82 m, which represents a 64% reduction in the error of the baseline model.

The result of the improved foot position extraction algorithm can be given in Table IV, demonstrating a reduction in distance error by roughly 0.4 m when compared to the original algorithm, and by approximately 1.4 m in comparison to the baseline model. This evidence also indicates that the improved foot position extraction algorithm proposed in this article has the capability to

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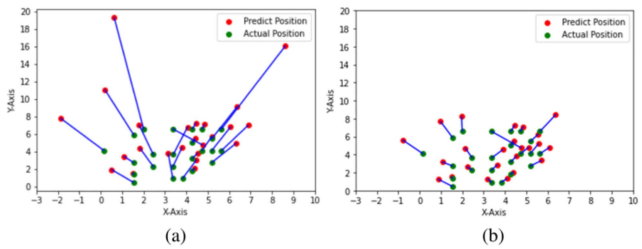


Fig. 14. Coordinates of the unimproved foot position extraction algorithm (a) and the improved foot position extraction algorithm and (b) after performing direct linear transformation.

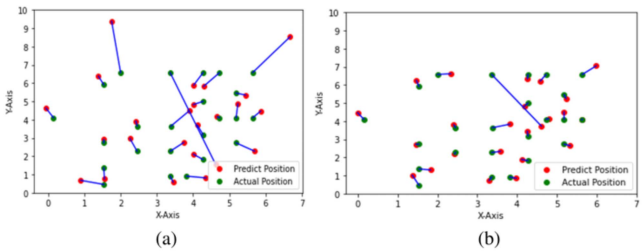


Fig. 15. Coordinates of the unimproved foot position extraction algorithm (a) and the improved foot position extraction algorithm (b) after performing direct linear transformation and applying machine learning techniques.

produce more accurate foot positions, thereby producing more precise environmental coordinates.

Fig. 14(a) displays the distance difference between the predicted and actual coordinates obtained through the 2-D-DLT and the unimproved foot position extraction algorithm. The error is smaller near the center but gradually increases around the field. This is due to the fact that in the unimproved algorithm, the extracted foot position is obscured in some cases, causing a significant discrepancy between the extracted foot position coordinates and the actual foot position coordinates, as well as lens distortion. In addition, there is no model available for correction. Therefore, this method has the largest error, approximately 2.2 m.

According to Fig. 14(b), the improved foot position extraction algorithm is able to produce predicted coordinates that are closer to the actual coordinates than Fig. 14(a), leading to a reduction in the number of coordinate points with excessive errors around the field. The overall prediction error decreased to approximately 0.8 m.

In Fig. 15(a), the errors in the area surrounding the field may increase slightly because the method does not utilize the improved foot position extraction algorithm. However, employing machine learning model can still significantly reduce the errors. Furthermore, there may be some visible differences between the predicted and actual coordinates, which can be attributed to the inaccurate foot position extraction algorithm. The distance error of this method is about 0.9 m.

In Fig. 15(b), the improved algorithm proposed for extracting foot positions resulted in extracted foot positions that are closer to the actual foot positions. Consequently, the predicted and actual coordinates are more closely aligned after using the machine learning model to correct the coordinates. It can be observed that most of the errors have been greatly reduced, with many predicted points being almost identical to the actual points; only a few points are predicted abnormally. This method has the lowest error among the four methods evaluated, with an error of only about 0.4 m. To provide a better understanding of the error statistics of this method, we also report the standard deviation and median, which are 0.59 m and 0.23 m, respectively.

Regarding the time complexity of proposed indoor positioning system, there are two time-consuming parts. One is our proposed image positioning method, which uses an algorithm based on the OpenPose model for real-time performance with low time complexity. This model uses a bottom-up approach to predict confidence maps for body parts, rather than the traditional top-down approach, which becomes less efficient as the number of people increases [10]. Compared to Mask RCNN [23], the method maintains the same runtime even in the case of a large number of people. After images are preprocessed with the OpenPose model, the resulting body postures are used as input for our efficient machine learning model, which has a time complexity of O(k), where k is the number of features in the data. The other time-consuming part is the receiving of Wi-Fi signals from the user's devices. This depends on the frequency with which the application transmits data to the server, which can be adjusted in the settings. In our experiment, the client sends data to the server every three seconds to update the location of the device in real-time and match it with the image positioning result.

Finally, we compare the results of our pose estimation-based positioning method with CamLoc, proposed by Cosma et al. [3]. In their study, they set up experimental environments and monitored the field from different angles using multiple cameras. The average distance error in five scenarios ranged between 0.38 and 0.58 m. In our experiment, the experimental field was relatively complex, with many computer screens, tables, and cabinets that could potentially obscure the body. Since it is uncommon for a real-world field to have more than one surveillance cameras, we only used one camera in our experiment. Our method, which incorporates the foot position extraction algorithm proposed in this article, achieved the same level of accuracy as CamLoc.

V. Conclusion

Owing to the popularity of smart cameras and Wi-Fi base stations in indoor environments, it is easier to capture the image and signal from public devices. Therefore, we proposed a novel indoor positioning system that takes advantage of Wi-Fi and image positioning. To improve the accuracy of Wi-Fi-based positioning, the Wi-Fi signal strengths of two frequency bands are collected as features of the machine learning models in order to adopt the variance of the environment and reduce the positioning error by about 0.25 m. In addition, we proposed the foot position extraction algorithm based on the OpenPose model and used only one surveillance camera to estimate the positions of the people in the image. The performance of the applied machine learning model improves the error by 64% compared to only using 2-D-DLT, and approximately 50% of the

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error can be reduced through the improved foot position extraction algorithm. Wi-Fi and image positioning could be used to not only match people's identities but also achieve highly accurate positioning. Nevertheless, the matching method proposed in this article has some limitations. Matching errors may occur when the distance between individuals is greater than the distance error of Wi-Fi-based positioning. As a result, the proposed method is only applicable when individuals are sparsely distributed. However, this issue can still be addressed. In the future, the matching accuracy can be improved as advancements are made in Wi-Fi-based positioning research.

REFERENCES

[1] C. Rose, J. Britt, J. Allen, and D. Bevly, "An integrated vehicle navigation system utilizing lane-detection and lateral position estimation systems in difficult environments for GPS," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 6, pp. 2615–2629, Dec. 2014.

[2] A. Poulose and D. S. Han, "UWB indoor localization using deep learning LSTM networks," *Appl. Sci.*, vol. 10, no. 18, Sep. 2020, Art. no. 6290.

[3] A. Cosma, I. E. Radoi, and V. Radu, "CamLoc: Pedestrian location estimation through body pose estimation on smart cameras," in *Proc. Int. Conf. Indoor Positioning Indoor Navig.*, 2019, pp. 1–8.

[4] H. Zhao, W. Cheng, N. Yang, S. Qiu, Z. Wang, and J. Wang, "Smartphone-based 3D indoor pedestrian positioning through multi-modal data fusion," *Sensors*, vol. 19, no. 20, Oct. 2019, Art. no. 4554.

[5] Z. Hao, J. Dang, W. Cai, and Y. Duan, "A multi-floor location method based on multi-sensor and WiFi fingerprint fusion," *IEEE Access*, vol. 8, pp. 223765–223781, 2020, doi: 10.1109/ACCESS.2020.3039394.

[6] "The growth in connected IoT devices is expected to generate 79.4ZB of data in 2025, according to a new IDC forecast," Businesswire.com, Framingham, Mass., 2019. Accessed: Aug. 10, 2022. [Online]. Available: https://www.businesswire.com/news/home/20190618005012/en/

[7] F. Liu et al., "Survey on WiFi-based indoor positioning techniques," *Inst. Eng. Technol. Commun.*, vol. 14, no. 9, pp. 1372–1383, Jun. 2020.

[8] K. Muthukrishnan, M. Lijding, and P. Havinga, "Towards smart surroundings: Enabling techniques and technologies for localization," in *Location- and Context-Awareness*. Berlin, Germay: Springer, 2005, pp. 350–362.

[9] Z. Liu, L. Zhang, Q. Liu, Y. Yin, L. Cheng, and R. Zimmermann, "Fusion of magnetic and visual sensors for indoor localization: Infrastructure-free and more effective," *IEEE Trans. Multimedia*, vol. 19, no. 4, pp. 874–888, Apr. 2017, doi: 10.1109/tmm.2016.2636750.

[10] X. Guo, N. Ansari, F. Hu, Y. Shao, N. R. Elikplim, and L. Li, "A survey on fusion-based indoor positioning," *IEEE Commun. Surveys Tut.*, vol. 22, no. 1, pp. 566–594, Jan.-Mar. 2020, doi: 10.1109/COMST.2019.2951036.

[11] A. Poulose, J. Kim, and D. S. Han, "A sensor fusion framework for indoor localization using smartphone sensors and WI-Fi RSSI measurements," *Appl. Sci. (Basel)*, vol. 9, no. 20, 2019, Art. no. 4379.

[12] T.-H. Tsai, C.-H. Chang, and S.-W. Chen, "Vision based indoor positioning for intelligent buildings," in *Proc. 2nd Int. Conf. Intell. Green Building Smart Grid*, 2016, pp. 1–4.

[13] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 580–587.

[14] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.

[15] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, "OpenPose: Realtime multi-person 2D pose estimation using part affinity fields," 2018.

[16] A. Poulose, J. Kim, and D. S. Han, "A sensor fusion framework for indoor localization using smartphone sensors and WI-Fi RSSI measurements," *Appl. Sci. (Basel)*, vol. 9, no. 20, 2019, Art. no. 4379.

[17] Y. I. Abdel-Aziz and H. M. Karara, "Direct linear transformation from comparator coordinates in close-range photogrammetry," 1971.

[18] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Computer Vision–ECCV 2014*, Berlin, Germay: Springer, 2014, pp. 740–755.

[19] F. Galton, "Regression towards mediocrity in hereditary stature," *J. Anthropolog. Inst. Great Britain Ireland*, vol. 15, pp. 246–263, 1886.

[20] M. E. Tipping, "Sparse Bayesian learning and the relevance vector machine," *J. Mach. Learn. Res.*, vol. 1, pp. 211–244, 2001.

[21] J. H. Friedman, "Greedy function approximation: A gradient boosting machine," *Ann. Statist.*, vol. 29, no. 5, pp. 1189–1232, 2001.

[22] N. A. M. Maung and W. Zaw, "Comparative study of RSS-based indoor positioning techniques on two different WI-Fi frequency bands," in *Proc. 17th Int. Conf. Elect. Eng. Electron., Comput., Telecommun. Inf. Technol.*, 2020, pp. 185–188.

[23] K. He, G. Gkioxari, P. Dollar, and R. Girshick, "Mask R-CNN," 2017, *arXiv:1703.06870*.



Han-Hsuan Cheng received the M.S. degree in electronic and computer engineering from the National Taiwan University of Science and Technology, Taipei, Taiwan, in 2022, with research in indoor positioning based on human pose estimation and Wi-Fi positioning.

He was a Software Engineer with the Chip Implementation CAD Department at the world's largest IC foundry, Taiwan Semiconductor Manufacturing Co. He is currently interested in deep learning and automating electrical design.



Jenq-Shiou Leu (Senior Member, IEEE) received the B.S. degree in mathematics and the M.S. degree in computer science and information engineering from the National Taiwan University, Taipei, Taiwan, in 1991 and 1993, respectively, and the Ph.D. degree on a part-time basis in computer science from the National TsingHua University, HsinChu, Taiwan, in 2006.

He was with Rising Star Technology, Taiwan, as an R&D Engineer, from 1995 to 1997, and worked with the telecommunication industry (Mobitai Communications and Taiwan Mobile) from 1997 to 2007, as an Assistant Manager. In February 2007, he was with the Department of Electronic and Computer Engineering at National Taiwan University of Science and Technology, as an Assistant Professor. From February 2011 to January 2014, he was an Associate Professor. Since February 2014, he has been a Professor. His research interests include mobile service and platform design and application development of computational intelligence.



Jin-Xian Liu received the B.S. degree in information management from the National Taipei University of Nursing and Health Sciences, Taipei, Taiwan, in 2021. He is currently working toward the M.S. degree in electronic and computer engineering with the National Taiwan University of Science and Technology, Taipei, Taiwan.

His research interests include machine learning and data analysis.

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